

Sharp logarithmic remainders for Ford's renewal recurrence

Spectral positivity, complete asymptotic expansions,
and sharp product bounds

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Abstract

Let $a_j = (j+1)\log(j+1) - j\log j - 1$, and let $g_0 = 1$, $g_n = \sum_{j=1}^n a_j g_{n-j}$. This recurrence enters the simplex-volume analysis of values of Euler's totient function. If $\sum_{j \geq 1} a_j \rho^j = 1$ and $\gamma = (\sum_{j \geq 1} j a_j \rho^j)^{-1}$, a published estimate bounds $g_n - \gamma \rho^{-n}$ absolutely by a constant. We prove the sharp remainder $g_n - \gamma \rho^{-n} \sim -1/(n^2 \log^2 n)$, and compute its complete expansion in inverse powers of $\log n$. An exact positive spectral measure proves strict complete monotonicity of $E_n = \gamma \rho^{-n} - g_n$ for $n \geq 1$, positive definiteness of all shifted Hankel matrices, and two exact sum rules. We also give a fully explicit finite-index two-sided bound. A local polylogarithm identity yields a second expansion in powers of $1/n$ whose coefficients retain their entire logarithmic dependence; this separates corrections that every fixed logarithmic truncation would hide. Finally we prove sharp bounds for the normalized product of the g_j , and invert the renewal-error law to estimate the index needed for a prescribed accuracy. The proofs are conventional mathematical proofs, supported by independent high-precision and spectral computations.

Status. This is a new research manuscript with internal independent proof checks, not an externally refereed or Lean-formalized publication. The exact claims and the limits of the priority search are recorded in Section 1. No theorem about the full totient counting function is assumed in these proofs.

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1 The problem and the precise improvement

The coefficients

$$a_j = (j+1)\log(j+1) - j\log j - 1 = \int_j^{j+1} \log x \, dx \quad (j \geq 1) \quad (1.1)$$

are positive and grow like $\log j$. Set

$$A(z) = \sum_{j \geq 1} a_j z^j, \quad g_0 = 1, \quad g_n = \sum_{j=1}^n a_j g_{n-j}, \quad G(z) = \frac{1}{1 - A(z)}. \quad (1.2)$$

The radius of convergence of A is one. On $[0, 1)$ it increases strictly from zero to infinity. Thus there is a unique $\rho \in (0, 1)$ with $A(\rho) = 1$. Throughout the paper,

$$\gamma = \frac{1}{\rho A'(\rho)}, \quad w_0 = \rho^{-1}, \quad E_n = \gamma \rho^{-n} - g_n \quad (n \geq 1). \quad (1.3)$$

Euler's constant is always γ_E , to distinguish it from the residue amplitude γ . All logarithms are natural. Numerical values, given only for orientation, are

$$\rho = 0.542598586098471021959384595779\dots, \quad \gamma = 0.323457534191380676574821548566\dots$$

1.1 Published baseline and repository connection

Ford's study of totients uses this recurrence in its simplex-volume estimates. Lemma 3.7 of the revised paper proves $|g_n - \gamma \rho^{-n}| \leq 5$; the following remark records that the errors for $n \geq 1$ are negative, increasing, and summable [1, Lemma 3.7 and Remark 1]. Ford attributes the simplified recurrence argument to a solution with Lau [2]. The present paper starts from those established facts as a comparison; it proves the new estimates directly.

The same recurrence is defined in the inspected `openai/math` totient manuscript, where a continuous counting scale contains

$$\mathcal{G}_m(B) = \frac{B^m}{m! \prod_{j=1}^m g_j}. \quad (1.4)$$

We use that definition and its motivation, without using the manuscript's claims about the arithmetic counting function [5]. The study also fits the asymptotic-expansion and inversion program in `ProveIt` [6]. Section 5 gives a direct, proved consequence for (1.4).

The priority review inspected Ford's full revised paper and the relevant repository source. The Ford–Lau item was identified bibliographically, but its full text was not accessible during preparation. Targeted searches did not locate the explicit remainder expansion proved here. This supports a precise claim of a refinement of the inspected published estimate; it does not establish worldwide first-publication priority. Spectral measures, Hausdorff moments, renewal generating functions, and polylogarithm expansions are classical tools.

1.2 Main conclusions at a glance

Theorem 1.1 (Sharp remainder and all logarithmic orders). *For every fixed integer $K \geq 0$, as $n \rightarrow \infty$,*

$$E_n = \frac{1}{n^2} \left\{ \sum_{k=0}^K \frac{c_k}{(\log n)^{k+2}} + O_K((\log n)^{-K-3}) \right\}, \quad (1.5)$$

where

$$c_k = (k+1)! [u^k] \frac{(1+u)e^{\gamma_E u}}{\Gamma(1-u)}. \quad (1.6)$$

In particular,

$$\begin{aligned} c_0 &= 1, & c_1 &= 2, & c_2 &= -\frac{\pi^2}{2}, \\ c_3 &= -2\pi^2 - 8\zeta(3), & c_4 &= \frac{\pi^4}{12} - 40\zeta(3). \end{aligned} \quad (1.7)$$

Moreover, $E_n > 0$ for every $n \geq 1$, and for every $n \geq 4096$,

$$\frac{1}{200n(n+1)\log^2 n} \leq E_n \leq \frac{10}{n(n+1)\log^2 n}. \quad (1.8)$$

Thus the formerly bounded absolute error actually tends to zero, with an exact leading constant and an optimal power of n and $\log n$. After just the first correction,

$$g_n = \gamma\rho^{-n} - \frac{1}{n^2 \log^2 n} + O\left(\frac{1}{n^2 \log^3 n}\right).$$

The new O -term is itself sharp: its leading coefficient is -2 . The more conservative constants in (1.8) provide a statement valid from a specified finite index, independently of any numerical calculation of ρ .

Theorem 1.2 (Exact moment structure). *There is an explicit positive continuous function ℓ on $(0, 1)$ such that*

$$E_n = \int_0^1 t^{n-1} \ell(t) dt \quad (n \geq 1). \quad (1.9)$$

Consequently $(-1)^r \Delta^r E_n > 0$ for all $n \geq 1$, $r \geq 0$, and every matrix $(E_{a+i+j})_{0 \leq i, j \leq d}$ with $a \geq 1$ is positive definite. The exact sum rules are

$$\sum_{n \geq 1} E_n = 1 - \frac{\gamma}{1-\rho}, \quad \sum_{n \geq 1} n E_n = \frac{\gamma\rho}{(1-\rho)^2}. \quad (1.10)$$

The restriction $n \geq 1$ matters: $g_0 - \gamma = 1 - \gamma > 0$ has the opposite sign. The proof explicitly checks both endpoints of the spectral measure and excludes an unaccounted singular component.

1.3 A second, finer expansion

A fixed truncation of (1.5) has an error much larger than n^{-3} . To resolve powers of $1/n$ as well, define the ordinary convergent integrals

$$F_a(L) = \int_0^\infty \frac{y^{a-1} e^{-y}}{(L - \log y - \gamma_E)^2 + \pi^2} dy \quad (a > 0, L \in \mathbb{R}), \quad (1.11)$$

and put

$$p_2 = \zeta'(-1) - \frac{1}{12}, \quad p_3 = \zeta'(-2). \quad (1.12)$$

Primes on F_a mean differentiation in L .

Theorem 1.3 (Expansion retaining all logarithmic dependence). *Writing $L = \log n$, one has*

$$\begin{aligned} E_n &= \frac{F_2(L)}{n^2} + \frac{F_3(L)}{2n^3} + \frac{1}{n^4} \left\{ \frac{F_4(L)}{12} - p_2 F_4'(L) \right\} \\ &\quad - \frac{1}{n^5} \left(\frac{p_2}{2} + p_3 \right) F_5'(L) + O\left(\frac{1}{n^6 \log^2 n}\right). \end{aligned} \quad (1.13)$$

More generally, an explicit recursion generates the coefficient of every power n^{-r-2} , with remainder $O_R(n^{-R-3} \log^{-2} n)$ after powers through n^{-R-2} .

In (1.13), the F_a are retained as integrals. Replacing them by a fixed number of inverse-logarithmic terms would destroy the displayed precision. This distinction is essential when using the result as an asymptotic expansion in more than one scale.

1.4 What these theorems do and do not settle

The main theorem improves a precise analytic component of the totient method. It also supplies exact inequalities for the entire recurrence, not merely its tail. The product theorem below determines the first relative correction to (1.4); the inverse theorem quantifies convergence of the associated renewal process.

These conclusions do not, by themselves, improve the error term in an asymptotic formula for the number of distinct totients. The arithmetic steps that connect a volume calculation to a count have separate errors. The present argument neither verifies nor requires the larger claims in the repository manuscript. This boundary keeps the strengthened estimate usable as an independently checkable lemma.

2 A positive spectral representation of the entire remainder

Put

$$a_j = \int_j^{j+1} \log x \, dx \quad (j \geq 1), \quad g_0 = 1, \quad g_n = \sum_{j=1}^n a_j g_{n-j},$$

and write $A(z) = \sum_{j \geq 1} a_j z^j$ and $G(z) = \sum_{n \geq 0} g_n z^n = 1/(1 - A(z))$. Let $\rho \in (0, 1)$ be the unique solution of $A(\rho) = 1$, put $w_0 = \rho^{-1}$, and set

$$\gamma = \frac{1}{\rho A'(\rho)}, \quad h(t) = \frac{(1-t)^2}{\log^2 t} \quad (0 < t < 1), \quad c = 2 \log 2.$$

The continuous extensions are $h(0) = 0$ and $h(1) = 1$. The constant γ is the residue amplitude; Euler's constant is denoted below by γ_E .

2.1 Elementary integral identities

The identity

$$\frac{1-t}{-\log t} = \int_0^1 t^u \, du$$

gives the useful representation

$$h(t) = \int_0^1 \int_0^1 t^{u+v} \, du \, dv. \quad (2.1)$$

In particular, Jensen's inequality and $0 < t < 1$ imply

$$t \leq h(t) \leq 1. \quad (2.2)$$

Direct differentiation of the formula for a_j , followed by the elementary identity $\log(v/u) = \int_0^\infty (e^{-ux} - e^{-vx}) \, dx/x$, gives

$$a_{j+1} - a_j = \int_0^1 t^{j-1} h(t) \, dt \quad (j \geq 1). \quad (2.3)$$

For completeness, one may also integrate the right side of (2.1) first in t : it equals $\int_0^1 \int_0^1 (j+u+v)^{-1} \, du \, dv$, and the two elementary integrations yield $(j+2) \log(j+2) - 2(j+1) \log(j+1) + j \log j$. Consequently

$$(1-z)A(z) = a_1 z + z^2 \int_0^1 \frac{h(t)}{1-tz} \, dt, \quad a_1 = c - 1.$$

Define, off $[0, 1]$,

$$J(w) = w - c - \int_0^1 \frac{h(t)}{w - t} dt. \quad (2.4)$$

Then initially for large $|w|$, and thereafter by analytic continuation,

$$G(1/w) = \frac{w - 1}{J(w)}. \quad (2.5)$$

Finally,

$$\int_0^1 \frac{h(t)}{t} dt = \int_0^\infty \frac{(1 - e^{-x})^2}{x^2} dx = 2 \log 2 = c. \quad (2.6)$$

Integration by parts reduces the last integral to $2 \int_0^\infty (e^{-x} - e^{-2x}) dx/x = 2 \log 2$.

2.2 A bounded self-adjoint operator

Work in the Hilbert space $\mathcal{H} = \mathbb{C} \oplus L^2((0, 1), h(t) dt)$ and define

$$B(x, f) = \left(cx + \int_0^1 f(t)h(t) dt, x + tf(t) \right). \quad (2.7)$$

This is a bounded self-adjoint operator: multiplication by t has norm at most one and the off-diagonal map is the bounded rank-one map induced by the constant function. Moreover,

$$\langle B(x, f), (x, f) \rangle = \int_0^1 th(t) \left| f(t) + \frac{x}{t} \right|^2 dt \geq 0. \quad (2.8)$$

The displayed square is integrable because $\int h(t)/t dt = c < \infty$; expanding it gives the preceding quadratic form exactly. Thus B is positive.

Let $e_0 = (1, 0)$. Solving $(w - B)(x, f) = e_0$ by its second component gives $f(t) = x/(w - t)$ and hence

$$\langle (w - B)^{-1}e_0, e_0 \rangle = \frac{1}{J(w)}. \quad (2.9)$$

By the bounded self-adjoint spectral theorem [4] there is a positive probability measure σ , supported in the bounded spectrum of B , such that

$$\frac{1}{J(w)} = \int \frac{\sigma(dt)}{w - t}. \quad (2.10)$$

There is exactly one spectral point above one. Indeed, on $(1, \infty)$,

$$J'(w) = 1 + \int_0^1 \frac{h(t)}{(w - t)^2} dt > 0,$$

whereas $J(w) \rightarrow -\infty$ as $w \downarrow 1$ and $J(w) \rightarrow \infty$ as $w \rightarrow \infty$. The unique zero is $w_0 = \rho^{-1}$. The Schur-complement calculation shows that every real $w \notin [0, 1]$ with $J(w) \neq 0$ is in the resolvent set. Positivity excludes negative spectrum. Thus

$$\text{supp } \sigma \subseteq [0, 1] \cup \{w_0\}.$$

The mass at w_0 is

$$b_0 = \sigma(\{w_0\}) = \frac{1}{J'(w_0)}, \quad \gamma = \frac{w_0 - 1}{w_0 J'(w_0)} = (1 - \rho)b_0. \quad (2.11)$$

The final equality can either be read from the residue in (2.5) or checked by differentiating that identity.

2.3 No hidden singular measure

For $0 < t < 1$, put

$$R(t) = t - c - \text{PV} \int_0^1 \frac{h(s)}{t-s} ds, \quad (2.12)$$

$$J_+(t) = R(t) + i\pi h(t). \quad (2.13)$$

The sign of the imaginary part follows from the upper-half-plane boundary value of $(w-s)^{-1}$. Since h is smooth and positive on compact subintervals of $(0, 1)$, the principal value and the boundary values are continuous there; the convergence from the upper half-plane is uniform on every smaller compact interval. Because $|J_+(t)| \geq \pi h(t) > 0$, Stieltjes inversion applied on these intervals proves absolute continuity, with density

$$\frac{d\sigma}{dt} = \frac{h(t)}{|J_+(t)|^2} \quad (0 < t < 1). \quad (2.14)$$

For clarity, this also excludes singular continuous mass in the open interval: inversion gives the measure of each compact interval as the integral of the continuous boundary density; such intervals exhaust $(0, 1)$.

Only endpoint atoms remain to be checked. From (2.6), for $\varepsilon > 0$,

$$J(-\varepsilon) = -\varepsilon \left(1 + \int_0^1 \frac{h(t)}{t(t+\varepsilon)} dt \right).$$

Since $\int_0^1 h(t)/t^2 dt = \infty$,

$$\sigma(\{0\}) = \lim_{\varepsilon \downarrow 0} \frac{-\varepsilon}{J(-\varepsilon)} = 0. \quad (2.15)$$

The divergence follows immediately from $h(t) \sim 1/\log^2 t$ at zero. At the other endpoint, $J(1+\varepsilon) \rightarrow -\infty$, so

$$\sigma(\{1\}) = \lim_{\varepsilon \downarrow 0} \frac{\varepsilon}{J(1+\varepsilon)} = 0. \quad (2.16)$$

The isolated atom at w_0 contributes zero to this latter limit. These arguments account for the entire measure:

$$\sigma = b_0 \delta_{w_0} + \frac{h(t)}{|J_+(t)|^2} \mathbf{1}_{(0,1)}(t) dt. \quad (2.17)$$

2.4 Exact positive remainder and its consequences

Since σ has total mass one, equations (2.5) and (2.10) give

$$G(1/w) - 1 = (w-1) \int \frac{\sigma(dt)}{w-t} - 1 = \int \frac{t-1}{w-t} \sigma(dt).$$

Expanding at infinity is justified by bounded spectral support. For every integer $n \geq 1$,

$$E_n := \gamma \rho^{-n} - g_n = \int_0^1 t^{n-1} \ell(t) dt > 0, \quad \ell(t) = \frac{(1-t)h(t)}{|J_+(t)|^2}. \quad (2.18)$$

This indexing is essential. The expression $\int t^n k(t) dt$ with $k = \ell/t$ is equivalent for $n \geq 1$, but it must not be extended to $n = 0$: the corresponding density is not integrable at zero. Indeed $J(-\varepsilon) \rightarrow 0^-$ by dominated convergence in (2.4) and (2.6), so $G(-1/\varepsilon) - 1 \rightarrow +\infty$ in (2.5).

Its Cauchy representation is a bounded isolated pole term plus $\int_0^1 \ell(t)/(t + \varepsilon) dt$; monotone convergence proves $\int_0^1 \ell(t)/t dt = \infty$. The exact generating-function identity is

$$G(z) = 1 + \frac{\gamma \rho^{-1} z}{1 - z/\rho} - z \int_0^1 \frac{\ell(t)}{1 - tz} dt. \quad (2.19)$$

The representation proves strict complete monotonicity:

$$(-1)^k \Delta^k E_n = \int_0^1 t^{n-1} (1-t)^k \ell(t) dt > 0 \quad (n \geq 1, k \geq 0), \quad (2.20)$$

where $\Delta E_n = E_{n+1} - E_n$. Every finite Hankel matrix $(E_{r+i+j})_{0 \leq i, j \leq m}$ is positive definite when $r \geq 1$: for any nonzero real polynomial $P(t) = \sum_{i=0}^m c_i t^i$,

$$\sum_{i, j=0}^m c_i c_j E_{r+i+j} = \int_0^1 t^{r-1} P(t)^2 \ell(t) dt > 0.$$

In particular $E_{n+1}^2 < E_n E_{n+2}$. These are exact finite- n statements, independent of any asymptotic expansion.

Several exact sum rules are also available. Comparing the coefficient of $1/w$ gives

$$\int_0^1 \ell(t) dt = \gamma \rho^{-1} - a_1 < 1. \quad (2.21)$$

Monotone convergence and the total mass of σ give

$$\sum_{n=1}^{\infty} E_n = \int_0^1 \frac{\ell(t)}{1-t} dt = 1 - b_0 = 1 - \frac{\gamma}{1-\rho}. \quad (2.22)$$

Letting $w \downarrow 1$ in (2.10) shows that $\int_0^1 (1-t)^{-1} \sigma(dt) = b_0/(w_0 - 1)$: the continuous part increases monotonically and the left side $1/J(w)$ tends to zero, while the isolated-pole term tends to $-b_0/(w_0 - 1)$. Consequently

$$\sum_{n=1}^{\infty} n E_n = \int_0^1 \frac{\ell(t)}{(1-t)^2} dt = \frac{b_0}{w_0 - 1} = \frac{\gamma \rho}{(1-\rho)^2}. \quad (2.23)$$

3 Explicit finite-index bounds and the sharp leading scale

We next prove actual numerical constants without needing certified quadrature for the spectral density. They are deliberately conservative; the exact integral (2.18) supports much tighter numerical enclosures.

3.1 The logarithmic endpoint constant

Define

$$C = \int_0^1 \frac{h(s) - 1}{1-s} ds.$$

With $s = e^{-x}$ this becomes

$$C = \int_0^{\infty} \left(\frac{e^{-x} - e^{-2x}}{x^2} - \frac{1}{e^x - 1} \right) dx.$$

Writing $E_1(a) = \int_a^\infty e^{-x} dx/x$, integration by parts after a cutoff ε gives

$$C = \lim_{\varepsilon \downarrow 0} \left\{ \frac{e^{-\varepsilon} - e^{-2\varepsilon}}{\varepsilon} - E_1(\varepsilon) + 2E_1(2\varepsilon) + \log(1 - e^{-\varepsilon}) \right\}.$$

The standard integral definition $\gamma_E = -\lim_{\varepsilon \downarrow 0} (E_1(\varepsilon) + \log \varepsilon)$ therefore yields

$$C = 1 - 2\log 2 - \gamma_E. \quad (3.1)$$

Only $0 < \gamma_E < 1$ is needed for the explicit bounds below. These inequalities follow, for example, from

$$\gamma_E = \int_0^1 \frac{1 - e^{-x}}{x} dx - \int_1^\infty \frac{e^{-x}}{x} dx.$$

Put $r(s) = (h(s) - 1)/(1 - s)$ and extend it at one by $r(1) = -1$. By (2.2), $-1 \leq r(s) \leq 0$. If $t = 1 - u$, then a direct decomposition $h(s) = 1 + (1 - s)r(s)$ gives

$$R(1 - u) = \log u + \gamma_E + \eta(u), \quad \eta(u) = -u - \log(1 - u) - u \operatorname{PV} \int_0^1 \frac{r(s)}{1 - u - s} ds. \quad (3.2)$$

Here R is the real part in (2.13).

We record explicit estimates for the last principal value. From (2.1), for $s \in [1/4, 1]$, $|h''(s)| \leq 32$: indeed $|q(q - 1)| \leq 2$ for $0 \leq q \leq 2$ and $s^{q-2} \leq 16$. Since r is the negative average of h' over $[s, 1]$, this implies $|r'(s)| \leq 16$ on that interval. For $t \in [1/2, 1)$,

$$\operatorname{PV} \int_0^1 \frac{r(s)}{t - s} ds = r(t) \log \frac{t}{1 - t} + \int_0^1 \frac{r(s) - r(t)}{t - s} ds.$$

The integral over $[0, 1/4]$ has absolute value at most one; the remaining part has absolute value at most $16(3/4) = 12$. Therefore, with $L = \log(1/u)$ and $0 < u \leq 1/2$,

$$|\eta(u)| \leq u(L + 14). \quad (3.3)$$

We used $0 \leq -u - \log(1 - u) \leq u^2$ for $u \leq 1/2$. In particular,

$$J_+(1 - u) = \log u + \gamma_E + i\pi + O(u \log(1/u)), \quad \ell(1 - u) \sim \frac{u}{\log^2(1/u)}. \quad (3.4)$$

3.2 Two-sided density bounds with actual constants

For $0 < u \leq e^{-4}$, $L \geq 4$ and $u(L + 14) \leq 18e^{-4} < 1/3$. Using $0 < \gamma_E < 1$, (3.3) implies

$$\frac{2}{3}L \leq |R(1 - u)| \leq \frac{4}{3}L.$$

Also $h(1 - u) \geq 1 - u > 1/2$, $h(1 - u) \leq 1$, and $\pi h(1 - u) \leq \pi < L$. Hence

$$\boxed{\frac{9}{50} \frac{u}{\log^2(1/u)} \leq \ell(1 - u) \leq \frac{9}{4} \frac{u}{\log^2(1/u)} \quad (0 < u \leq e^{-4}).} \quad (3.5)$$

3.3 A fully explicit finite-index theorem

For every integer $n \geq 4096$,

$$\boxed{\frac{1}{200n(n+1)\log^2 n} \leq \gamma \rho^{-n} - g_n \leq \frac{10}{n(n+1)\log^2 n}.} \quad (3.6)$$

To prove the upper bound, split the integral in (2.18) at $u = n^{-1/2} \leq e^{-4}$. On the part $u \leq n^{-1/2}$, $\log(1/u) \geq \frac{1}{2} \log n$, so

$$\int_0^{n^{-1/2}} (1-u)^{n-1} \ell(1-u) du \leq \frac{9}{\log^2 n} \int_0^1 (1-u)^{n-1} u du = \frac{9}{n(n+1) \log^2 n}.$$

On the complement use $\int \ell \leq 1$, a consequence of the probability measure σ :

$$\int_{n^{-1/2}}^1 (1-u)^{n-1} \ell(1-u) du \leq e^{-\sqrt{n}+1/\sqrt{n}} \leq \frac{1}{n(n+1) \log^2 n}.$$

For the last elementary inequality set $r = \sqrt{n} \geq 64$. The difference between the logarithms of the two sides, with signs reversed, is

$$r - r^{-1} - 4 \log r - \log(1 + r^{-2}) - 2 \log(2 \log r).$$

It is positive at $r = 64$ (use $\log 64 < 5$ and $\log 10 < 3$), and its derivative is positive for $r \geq 64$.

For the lower bound restrict to $1/(2n) \leq u \leq 1/n$. On that interval, $(1-u)^{n-1} \geq e^{-1} > 1/3$ and $\log(1/u) \leq \log(2n) \leq 2 \log n$. Equation (3.5) gives

$$E_n \geq \frac{9}{600 \log^2 n} \int_{1/(2n)}^{1/n} u du = \frac{9}{1600 n^2 \log^2 n} \geq \frac{1}{200 n(n+1) \log^2 n}.$$

No numerical root approximation, numerical principal value, or numerical integration is used in (3.6).

A weaker bound valid at every $n \geq 2$ follows from $|J_+(t)|^2 \geq \pi^2 h(t)^2$ and $h(t) \geq t$:

$$0 < E_n \leq \frac{1}{\pi^2 n(n-1)} \quad (n \geq 2). \quad (3.7)$$

3.4 Sharp asymptotic constant

The same estimates give

$$E_n \sim \frac{1}{n^2 \log^2 n}. \quad (3.8)$$

Here are details avoiding an implicit Tauberian hypothesis. The part $u \geq n^{-1/2}$ is negligible after multiplication by $n^2 \log^2 n$, as just shown. In the remaining integral put $v = nu$. For each fixed $v > 0$, (3.4) gives

$$n^2 \log^2 n (1-v/n)^{n-1} \ell(1-v/n) \frac{dv}{n} \longrightarrow v e^{-v} dv.$$

For $v \leq \sqrt{n}$, (3.5) bounds the integrand by $9ve^{-v/2}$ for all sufficiently large n . Dominated convergence applies, and $\int_0^\infty v e^{-v} dv = 1$.

Thus the error has a fixed negative sign in $g_n - \gamma \rho^{-n}$, its absolute order $n^{-2}(\log n)^{-2}$ cannot be improved, and its entire finite-index shape is constrained by a positive Hausdorff moment representation.

4 Complete expansions in two asymptotic scales

The positive measure resolves global singularities and signs. We now analyze its endpoint with enough precision to recover every inverse-logarithmic coefficient and every fixed algebraic order. The first expansion keeps only powers of $1/\log n$; the second retains these logarithmic effects inside convergent integral functions before expanding in $1/n$.

We use J , h , and ℓ from Section 2. The boundary value is $J_+(t) = R(t) + i\pi h(t)$. All truncation indices and all parameters $a > 0$ below are fixed while n or L tends to infinity.

4.1 The exact analytic endpoint correction

The stronger identity is

$$\frac{J(e^{-s} + i0)}{h(e^{-s})} = \log s + \gamma_E + P(s) + i\pi, \quad 0 < s < 2\pi, \quad (4.1)$$

where the real-analytic function P has the convergent expansion

$$P(s) = \frac{s^2}{4 \sinh^2(s/2)} - 1 + \sum_{k=0}^{\infty} \zeta'(-1-k) \frac{s^{k+2}}{k!}. \quad (4.2)$$

The assertion for the whole interval $(0, 2\pi)$ means continuation of the convergent Taylor expression there. Its local use near zero is all that is needed below. In particular,

$$P(s) = p_2 s^2 + p_3 s^3 + p_4 s^4 + O(s^5), \quad p_2 = \zeta'(-1) - \frac{1}{12}, \quad p_3 = \zeta'(-2), \quad p_4 = \frac{1}{2} \zeta'(-3) + \frac{1}{240}.$$

Here is a derivation fixing the signs. Define

$$\mathcal{B}(z) = \sum_{m \geq 1} m \log m z^m = -\partial_\nu \text{Li}_\nu(z)|_{\nu=-1}.$$

The coefficients of $A(z) = \sum_{j \geq 1} a_j z^j$ give

$$A(z) = (z^{-1} - 1)\mathcal{B}(z) - \frac{z}{1-z}, \quad (1-z)(1-A(z)) = 1 - \frac{(1-z)^2}{z} \mathcal{B}(z).$$

Differentiate the classical convergent polylogarithm expansion [3, 25.12.12]

$$\text{Li}_\nu(e^{-u}) = \Gamma(1-\nu)u^{\nu-1} + \sum_{k \geq 0} \zeta(\nu-k) \frac{(-u)^k}{k!}, \quad |u| < 2\pi,$$

at $\nu = -1$, using the principal logarithm away from the negative real axis. It gives

$$\mathcal{B}(e^{-u}) = \frac{1 - \gamma_E - \log u}{u^2} - \sum_{k \geq 0} \zeta'(-1-k) \frac{(-u)^k}{k!}.$$

For $w = e^u$ one has $J(w) = w(1-w^{-1})(1-A(w^{-1}))$. Substitute the last display and then let $u \rightarrow -s + i0$. Since $\log(-s + i0) = \log s + i\pi$ and

$$h(e^{-s}) = e^{-s} \frac{4 \sinh^2(s/2)}{s^2},$$

the result is (4.1)–(4.2). The lower boundary has $-i\pi$. The positive sign on the upper boundary also agrees with the direct Sokhotski–Plemelj formula for J .

4.2 A Laplace form and the correct scale of the first correction

Let $L = \log n$ and, for $a > 0$, define

$$F_a(L) = \int_0^\infty \frac{y^{a-1} e^{-y}}{(L - \log y - \gamma_E)^2 + \pi^2} dy. \quad (4.3)$$

These functions are positive, finite, and differentiable to every order on the real axis. The denominator is bounded below by π^2 . In (2.18) the change of variables $t = e^{-s}$ is especially useful:

$$E_n = \int_0^\infty e^{-ns} \ell(e^{-s}) ds.$$

For sufficiently small positive s , (4.1) yields the exact local density expression

$$\ell(e^{-s}) = \frac{s^2}{1 - e^{-s}} \frac{1}{(\log s + \gamma_E + P(s))^2 + \pi^2}. \quad (4.4)$$

The part of the original moment integral with $t \leq e^{-\delta}$ is $O(e^{-(n-1)\delta})$, since ℓ is integrable. Thus only a fixed neighbourhood of $s = 0$ enters any of the algebraic orders below.

The expansion $P(s) = O(s^2)$ in (4.1) gives

$$\ell(e^{-s}) = \frac{s + s^2/2}{(\log s + \gamma_E)^2 + \pi^2} + O\left(\frac{s^3}{(\log s + \gamma_E)^2 + \pi^2}\right).$$

Consequently

$$E_n = \frac{F_2(L)}{n^2} + \frac{F_3(L)}{2n^3} + O\left(\frac{1}{n^4 L^2}\right). \quad (4.5)$$

In particular the first relative correction in powers of $1/n$ is positive.

For comparison, the direct t -variable model satisfies the weaker but useful bound

$$E_n = \int_0^1 \frac{t^{n-1}(1-t)}{(\log(1-t) + \gamma_E)^2 + \pi^2} dt + O\left(\frac{1}{n^3 \log^2 n}\right).$$

Its error has a genuine $n^{-3} \log^{-2} n$ contribution; changing to the Laplace variable incorporates the corresponding first correction cleanly.

4.3 Every fixed algebraic order, retaining the logarithmic dependence

Put

$$f(V) = \frac{1}{V^2 + \pi^2}, \quad b(s) = \frac{s}{1 - e^{-s}}, \quad b(s)f(V + P(s)) = \sum_{r \geq 0} Q_r(V) s^r.$$

The last identity defines Q_r by Taylor expansion in s while V is held fixed. Each Q_r is a finite linear combination of derivatives of f , with constants obtained from the Bernoulli coefficients of b and the Taylor coefficients of P . Explicitly,

$$\begin{aligned} Q_0 &= f, & Q_1 &= \tfrac{1}{2}f, \\ Q_2 &= \tfrac{1}{12}f + p_2 f', & Q_3 &= (p_2/2 + p_3)f', \\ Q_4 &= -\tfrac{1}{720}f + (p_2/12 + p_3/2 + p_4)f' + \tfrac{1}{2}p_2^2 f''. \end{aligned}$$

Define the coefficient functions

$$\Psi_r(L) = \int_0^\infty y^{r+1} e^{-y} Q_r(\log y - L + \gamma_E) dy.$$

Then, for every fixed integer $R \geq 0$,

$$E_n = \sum_{r=0}^R \frac{\Psi_r(\log n)}{n^{r+2}} + O_R\left(\frac{1}{n^{R+3} \log^2 n}\right). \quad (4.6)$$

Here is a uniform remainder proof. Choose a small fixed circle $|\xi| = r_0 < 2\pi$ on which b and P are analytic, and let $M = \sup_{|\xi| \leq r_0} |P(\xi)|$. For real $|V| > 2(M + \pi + 1)$, the function $b(\xi)f(V + P(\xi))$ is analytic in that disk and is bounded by C/V^2 . Cauchy's estimate for its Taylor remainder, for $0 < s < r_0/2$, gives

$$b(s)f(V + P(s)) - \sum_{r=0}^R Q_r(V) s^r = O_R(s^{R+1}/V^2).$$

Choose δ small enough that this applies to $V = \log s + \gamma_E$ for $0 < s < \delta$. Multiplication by s in (4.4) makes the local density error $O_R(s^{R+2}/\log^2 s)$. For each fixed $m \geq 0$,

$$\int_0^\delta e^{-ns} \frac{s^m}{1 + \log^2 s} ds = O_m(n^{-m-1} \log^{-2} n).$$

Indeed, on $s \leq n^{-1/2}$ the denominator is at least a constant times $\log^2 n$, and the remaining part is exponentially small in $\sqrt{n}$. This proves the remainder in (4.6). Finally, every Q_r is bounded on the real axis and decays as $O_r(V^{-2})$, so extending the coefficient integrals from $y < n\delta$ to $y < \infty$ adds only exponentially small errors.

With the convention that a prime on F_a means differentiation in L , the first four power sectors are therefore

$$\begin{aligned} E_n = & \frac{F_2(L)}{n^2} + \frac{F_3(L)}{2n^3} + \frac{1}{n^4} \left\{ \frac{F_4(L)}{12} - p_2 F_4'(L) \right\} \\ & - \frac{p_2/2 + p_3}{n^5} F_5'(L) + O\left(\frac{1}{n^6 L^2}\right). \end{aligned} \quad (4.7)$$

The minus signs in the derivative terms are forced by $V = \log y - L + \gamma_E$; f is even but f' is odd. The n^{-6} coefficient, if wanted, is

$$-\frac{F_6}{720} - (p_2/12 + p_3/2 + p_4)F_6' + \frac{p_2^2}{2}F_6''.$$

It can be included with an $O(n^{-7}L^{-2})$ remainder by taking $R = 4$.

Essential scope distinction. In (4.6), the functions Ψ_r (or the F_a in (4.7)) remain intact. Truncating the inverse-logarithmic expansion of the leading sector at any fixed order produces an error larger than every later fixed power of $1/n$. A finite rectangular double sum does not inherit the remainder in (4.6).

4.4 Every fixed inverse-logarithmic order

For $k \geq 0$ put

$$q_k(U) = \frac{(U + i\pi)^{k+1} - (U - i\pi)^{k+1}}{2i\pi} = \sum_{j=0}^{\lfloor k/2 \rfloor} (-1)^j \binom{k+1}{2j+1} \pi^{2j} U^{k-2j}.$$

For every fixed $a > 0$ and $K \geq 0$,

$$F_a(L) = \Gamma(a) \sum_{k=0}^K \frac{c_{a,k}}{L^{k+2}} + O_{a,K}(L^{-K-3}), \quad L \rightarrow \infty, \quad (4.8)$$

where

$$c_{a,k} = \frac{1}{\Gamma(a)} \int_0^\infty y^{a-1} e^{-y} q_k(\log y + \gamma_E) dy.$$

To prove the remainder, split the integral into $y < e^{-L/2}$, $e^{-L/2} \leq y \leq e^{L/2}$, and $y > e^{L/2}$. The outer parts, including the tails of the proposed coefficients, are smaller than every fixed inverse power of L . On the middle part expand

$$\frac{1}{(L-U)^2 + \pi^2} = \frac{1}{2i\pi} \left\{ \frac{1}{L-U-i\pi} - \frac{1}{L-U+i\pi} \right\}.$$

For sufficiently large L , $|U \pm i\pi| \leq 3L/4$ there. The finite geometric expansion through the required order has a remainder bounded by $C_K L^{-K-3}(1 + |U|^{K+2})$. All logarithmic moments against $y^{a-1}e^{-y}$ are finite, proving (4.8).

The coefficient generating function has a particularly simple form:

$$\sum_{k \geq 0} \frac{c_{a,k}}{(k+1)!} z^k = \frac{\Gamma(a+z)}{\Gamma(a)} e^{\gamma_E z} \frac{\sin \pi z}{\pi z}. \quad (4.9)$$

This follows first for small z by integrating $\sum q_k(U) z^k / (k+1)! = e^{Uz} \sin(\pi z) / (\pi z)$. For a positive integer a the reflection formula rewrites it as

$$\sum_{k \geq 0} \frac{c_{a,k}}{(k+1)!} z^k = \left\{ \prod_{j=1}^{a-1} (1 + z/j) \right\} \frac{e^{\gamma_E z}}{\Gamma(1-z)}.$$

The right-hand side is entire. This is a coefficient-generation statement; it does not assert convergence of the asymptotic inverse-log series.

For the leading sector, $a = 2$, let $c_k = c_{2,k}$. Then

$$\sum_{k \geq 0} \frac{c_k}{(k+1)!} z^k = (1+z) \frac{e^{\gamma_E z}}{\Gamma(1-z)} = (1+z) \exp \left\{ - \sum_{m \geq 2} \frac{\zeta(m)}{m} z^m \right\}. \quad (4.10)$$

The displayed exponential identity holds in $|z| < 1$; the preceding gamma quotient gives its entire continuation. The first five coefficients are

$$c_0 = 1, \quad c_1 = 2, \quad c_2 = -\frac{\pi^2}{2}, \quad c_3 = -2\pi^2 - 8\zeta(3), \quad c_4 = \frac{\pi^4}{12} - 40\zeta(3).$$

For a recursive implementation set $b_0 = 1$, $b_1 = 0$ and

$$b_k = -\frac{1}{k} \sum_{m=2}^k \zeta(m) b_{k-m} \quad (k \geq 2), \quad c_k = (k+1)!(b_k + b_{k-1}),$$

where $b_{-1} = 0$. Thus, for every fixed K ,

$$g_n = \gamma \rho^{-n} - \frac{1}{n^2} \sum_{k=0}^K \frac{c_k}{\log^{k+2} n} + O_K \left(\frac{1}{n^2 \log^{K+3} n} \right). \quad (4.11)$$

In particular,

$$g_n - \gamma \rho^{-n} = -\frac{1}{n^2 \log^2 n} \left\{ 1 + \frac{2}{\log n} - \frac{\pi^2}{2 \log^2 n} - \frac{2\pi^2 + 8\zeta(3)}{\log^3 n} + O(\log^{-4} n) \right\}.$$

For the n^{-3} sector in (4.5), the normalized first coefficients are

$$\frac{1}{2} F_3(L) = \frac{1}{L^2} \left\{ 1 + \frac{3}{L} + \frac{3 - \pi^2/2}{L^2} - \frac{3\pi^2 + 8\zeta(3)}{L^3} + O(L^{-4}) \right\}.$$

Their generating function is $(1+z)(1+z/2)e^{\gamma_E z} / \Gamma(1-z)$.

5 Products and the continuous totient counting scale

The recurrence enters (1.4) multiplicatively, so an additive estimate should be transported through the product with its sign and its first correction intact. Define

$$\delta_n = \frac{\rho^n E_n}{\gamma}, \quad P_m = \prod_{j=1}^m \frac{g_j}{\gamma \rho^{-j}} = \prod_{j=1}^m (1 - \delta_j). \quad (5.1)$$

Positivity of g_j and the exact moment representation give $0 < \delta_j < 1$. Moreover,

$$0 < \delta_{j+1} < \rho \delta_j, \quad (5.2)$$

because $E_{j+1} < E_j$. Hence P_m decreases to a strictly positive constant

$$C_* := \prod_{j=1}^{\infty} (1 - \delta_j) \in (0, 1). \quad (5.3)$$

Strict positivity follows from convergence of $\sum \delta_j$ and the fact that none of the individual factors vanishes.

Theorem 5.1 (Sharp product tail). *For every fixed $K \geq 0$, set $n = m + 1$ and $L = \log n$. Then*

$$\log \frac{P_m}{C_*} = \frac{\rho^n}{\gamma(1 - \rho)n^2} \left\{ \sum_{k=0}^K \frac{c_k}{L^{k+2}} + O_K(L^{-K-3}) \right\}. \quad (5.4)$$

In particular,

$$P_m = C_* \left[1 + \frac{\rho^{m+1}}{\gamma(1 - \rho)(m + 1)^2 \log^2(m + 1)} \left(1 + \frac{2}{\log(m + 1)} + O(\log^{-2}(m + 1)) \right) \right]. \quad (5.5)$$

Proof. Write

$$T_m = \sum_{j=m+1}^{\infty} \delta_j, \quad R_m = \sum_{j=m+1}^{\infty} [-\log(1 - \delta_j) - \delta_j].$$

Then $\log(P_m/C_*) = T_m + R_m$. The elementary power series for $-\log(1 - x)$ and (5.2) give

$$0 \leq R_m \leq \frac{\delta_n^2}{2(1 - \delta_n)(1 - \rho^2)}, \quad n = m + 1. \quad (5.6)$$

Tonelli's theorem applied to the positive moment representation gives the exact identity

$$T_m = \frac{\rho^n}{\gamma} \int_0^1 \frac{t^{n-1} \ell(t)}{1 - \rho t} dt. \quad (5.7)$$

Put $\beta = \rho/(1 - \rho)$ and

$$D_{n,r} = (-1)^r \Delta^r E_n = \int_0^1 t^{n-1} (1 - t)^r \ell(t) dt. \quad (5.8)$$

Since $(1 - \rho t)^{-1} = (1 - \rho)^{-1} [1 - \beta(1 - t)/(1 + \beta(1 - t))]$,

$$\frac{\rho^n}{\gamma(1 - \rho)} (E_n - \beta D_{n,1}) \leq T_m \leq \frac{\rho^n E_n}{\gamma(1 - \rho)}. \quad (5.9)$$

The same endpoint argument that gives the main theorem yields $D_{n,1} = O(n^{-3} \log^{-2} n)$; a proof with all coefficients is given in Section 7. Thus replacing $(1 - \rho t)^{-1}$ by $(1 - \rho)^{-1}$ costs an extra factor $1/n$. For every fixed K , this error and (5.6) are absorbed by the last term of (5.4). Exponentiating proves (5.5). $\square$

5.1 Finite, computable two-sided bounds for the product constant

The integral proof gives more than an asymptotic notation. For $n = m + 1$, let

$$\begin{aligned} U_m &= \frac{\delta_n}{1 - \rho} + \frac{\delta_n^2}{2(1 - \delta_n)(1 - \rho^2)}, \\ L_m &= \max \left\{ \delta_n, \frac{\rho^n}{\gamma(1 - \rho)} (E_n - \beta D_{n,1}) \right\}. \end{aligned} \quad (5.10)$$

Every $m \geq 0$ satisfies

$$\boxed{P_m e^{-U_m} \leq C_* \leq P_m e^{-L_m}}. \quad (5.11)$$

The additional lower bound $T_m \geq \delta_n$ used here follows directly from its defining positive sum. These formulas involve just finitely many recurrence terms and the two scalar constants ρ, γ . They become certified numerical enclosures when those inputs are enclosed by outward-rounded interval arithmetic. The supplied high-precision computations are diagnostics, and are not described as interval certificates.

There is a systematic refinement. The exact finite geometric identity gives, for every $r \geq 1$,

$$\begin{aligned} T_m &= \frac{\rho^n}{\gamma(1 - \rho)} \sum_{j=0}^{r-1} (-\beta)^j D_{n,j} + \mathcal{R}_{m,r}, \\ 0 &< (-1)^r \mathcal{R}_{m,r} \leq \frac{\rho^n \beta^r}{\gamma(1 - \rho)} D_{n,r}. \end{aligned} \quad (5.12)$$

No convergence of an infinite geometric series with ratio β is being used: $\beta > 1$ is allowed. The signed remainder is the exact integral with factor $(1 + \beta(1 - t))^{-1}$. For each fixed r , $D_{n,r} = O_r(n^{-r-2} \log^{-2} n)$, so successive bounds resolve successive powers of $1/n$.

5.2 Effect on the counting scale

From the definition of P_m one obtains the exact algebraic identity

$$\mathcal{G}_m(B) = \frac{B^m \rho^{m(m+1)/2}}{m! \gamma^m C_*} \exp \left(-\log \frac{P_m}{C_*} \right). \quad (5.13)$$

It follows that the approximation obtained by replacing P_m with C_* is an upper approximation. Its relative correction has the sharp size

$$\frac{\rho^{m+1}}{\gamma(1 - \rho)(m + 1)^2 \log^2(m + 1)}. \quad (5.14)$$

This improves the product-tail estimate $O(\rho^m)$ obtainable from a bounded absolute recurrence error by the factor $m^{-2} \log^{-2} m$, and determines its leading constant and sign. It is a statement about the scale (1.4). It does not identify or eliminate independent arithmetic errors in a totient counting argument.

6 A renewal process and inversion of its convergence rate

Define a probability distribution on the positive integers by

$$p_j = a_j \rho^j \quad (j \geq 1). \quad (6.1)$$

Its total mass is $A(\rho) = 1$ and its mean is $\sum j p_j = 1/\gamma$. For independent positive integer-valued increments with this law, let u_n be the probability that a renewal occurs at time n , with $u_0 = 1$. Splitting according to the first increment gives $u_n = \sum_{j=1}^n p_j u_{n-j}$, so

$$u_n = \rho^n g_n, \quad \gamma - u_n = \rho^n E_n \quad (n \geq 1). \quad (6.2)$$

Thus the renewal probabilities increase strictly to γ from time one onward, and

$$\gamma - u_n = \frac{\rho^n}{n^2 \log^2 n} \left(1 + \frac{2}{\log n} - \frac{\pi^2}{2 \log^2 n} + O(\log^{-3} n) \right). \quad (6.3)$$

The exponential rate is exactly ρ : $\lim(\gamma - u_n)^{1/n} = \rho$. No bound $O(r^n)$ with $r < \rho$ can hold. The extra polynomial and logarithmic factors describe the behavior at this exact exponential rate.

The moment formula also extends the error to real indices:

$$E(x) = \int_0^1 t^{x-1} \ell(t) dt, \quad \delta(x) = \frac{\rho^x E(x)}{\gamma} \quad (x \geq 1). \quad (6.4)$$

It is continuous for $x \geq 1$, smooth for $x > 1$, and strictly decreasing to zero. All the asymptotic proofs above remain valid for real $x \rightarrow \infty$.

Theorem 6.1 (Inverse error law). *Let $0 < \varepsilon < \delta(1)$, and let x_ε be the unique real solution of $\delta(x_\varepsilon) = \varepsilon$. Write*

$$a = \log(1/\rho), \quad B = \log(1/\varepsilon), \quad T = \log B, \quad c = \log a.$$

Then as $\varepsilon \downarrow 0$,

$$\begin{aligned} ax_\varepsilon = B - 2T - 2\log T + 2c - \log \gamma + \frac{2c+2}{T} \\ + \frac{c^2 + 2c - \pi^2/2 - 2}{T^2} + O(T^{-3}). \end{aligned} \quad (6.5)$$

The least positive integer index with relative renewal error at most ε is exactly $\lceil x_\varepsilon \rceil$.

Proof. The logarithm of (1.5), including its first three coefficients, gives

$$B = ax + 2\log x + 2\log \log x + \log \gamma - \frac{2}{\log x} + \frac{\pi^2/2 + 2}{\log^2 x} + O(\log^{-3} x). \quad (6.6)$$

Its first-order balance implies $x \sim B/a$. Substituting the right side of (6.5), divided by a , gives

$$\log x = T - c + O(T/B), \quad \log \log x = \log T - \frac{c}{T} - \frac{c^2}{2T^2} + O(T^{-3} + B^{-1}).$$

Matching the coefficients of $1/T$ and $1/T^2$ in (6.6) gives $2c+2$ and $c^2 + 2c - \pi^2/2 - 2$, respectively. The discarded error T/B is $O(T^{-3})$. Finally the exact function $-\log \delta(x)$ has derivative

$$a - \frac{E'(x)}{E(x)} \geq a > 0,$$

because $E'(x) = \int t^{x-1} \log t \ell(t) dt < 0$. Therefore a residual $O(T^{-3})$ in the logarithmic equation changes its solution by at most $O(T^{-3})$. Strict monotonicity proves the statement about the least integer index. $\square$

The last sentence uses the exact real solution. Rounding a truncated asymptotic approximation requires an enclosure if that approximation is close to an integer. At the purely asymptotic level the integer index has an unavoidable $O(1)$ rounding uncertainty.

7 Finite differences and tails of the spectral remainder

The moment structure permits several extensions without a new complex contour argument. They also verify that the product estimates above use a quantitatively controlled error.

Proposition 7.1 (Every fixed finite-difference order). *For fixed integers $r, K \geq 0$,*

$$D_{n,r} = \frac{1}{n^{r+2}} \left\{ \sum_{k=0}^K \frac{d_{r,k}}{(\log n)^{k+2}} + O_{r,K}((\log n)^{-K-3}) \right\}, \quad (7.1)$$

where

$$d_{r,k} = (k+1)! [u^k] \frac{e^{\gamma_E u} \prod_{j=1}^{r+1} (j+u)}{\Gamma(1-u)}. \quad (7.2)$$

If $H_s = \sum_{j=1}^s j^{-1}$ and $H_s^{(2)} = \sum_{j=1}^s j^{-2}$, the first terms are

$$\begin{aligned} d_{r,0} &= (r+1)!, \\ d_{r,1} &= 2(r+1)! H_{r+1}, \\ d_{r,2} &= 3(r+1)! \left(H_{r+1}^2 - H_{r+1}^{(2)} - \zeta(2) \right). \end{aligned} \quad (7.3)$$

Proof. In (5.8), put $t = e^{-s}$. The local integrand is the one for E_n multiplied by $(1 - e^{-s})^r = s^r(1 + O_r(s))$. The leading logarithmic sector is consequently $n^{-r-2} F_{r+2}(\log n)$, while the discarded powers of s cost an extra factor $1/n$. The all-orders inverse-logarithm proof applies with $y^{r+1}e^{-y}$ in place of ye^{-y} . The reflection identity for Γ simplifies the coefficient generator to (7.2). Expanding its logarithm through degree two gives (7.3). $\square$

Two tails deserve separate attention. They follow from exact summation under the positive integral:

$$\begin{aligned} \sum_{j>N} E_j &= \int_0^1 \frac{t^N \ell(t)}{1-t} dt, \\ \sum_{j>N} j E_j &= \int_0^1 \left(\frac{(N+1)t^N}{1-t} + \frac{t^{N+1}}{(1-t)^2} \right) \ell(t) dt. \end{aligned} \quad (7.4)$$

In particular,

$$\begin{aligned} \sum_{j>N} E_j &= \frac{1}{N \log^2 N} \left(1 - \frac{\pi^2}{2 \log^2 N} + O(\log^{-3} N) \right), \\ \sum_{j>N} j E_j &= \frac{1}{\log N} + \frac{1}{\log^2 N} - \frac{\pi^2}{6 \log^3 N} + O(\log^{-4} N). \end{aligned} \quad (7.5)$$

For the first formula the coefficient of $1/\log N$ inside the parentheses vanishes: the generator is $e^{\gamma_E u}/\Gamma(1-u)$, without the factor $1+u$. Alternatively, integrate the expansion of E_x and bound the sum–integral difference by the first term, using monotonicity. For the second formula, termwise integration of $x^{-1}(\log x)^{-2}$ and its corrections gives the displayed constants. To justify the sum–integral error in the weighted case, put $f(x) = xE(x)$. The moment formula gives $E'(x) \leq 0$, and integration by parts, using $xE(x) \rightarrow 0$, yields

$$\int_N^\infty |f'(x)| dx \leq NE(N) + 2 \int_N^\infty E(x) dx = O(N^{-1} \log^{-2} N).$$

Comparing $f(j)$ with its integral over $[j-1, j]$ bounds the sum–integral error by this total variation. This does not require differentiating an asymptotic error term. Thus the weighted sum rule (1.10) converges only logarithmically, even though g_n itself grows exponentially.

8 A universality theorem for logarithmic moment increments

The main mechanism is reusable. The following sufficient conditions make its precise range transparent; no claim is made that they are minimal.

Theorem 8.1 (A class of logarithmic renewal recurrences). *Let h be continuous on $[0, 1]$, positive on $(0, 1]$, continuously differentiable on $(0, 1)$, and twice continuously differentiable near one. Suppose*

$$H = h(1) > 0, \quad I_0 = \int_0^1 \frac{h(t)}{t} dt < \infty.$$

Choose $c > 1$ with $c \geq I_0$. Define

$$a_1 = c - 1, \quad a_{j+1} - a_j = \int_0^1 t^{j-1} h(t) dt,$$

and define g_n, ρ, γ from these coefficients as in (1.2)–(1.3). Put

$$b = \frac{1 - c - \int_0^1 \frac{h(t) - H}{1 - t} dt}{H}. \quad (8.1)$$

Then $E_n = \gamma \rho^{-n} - g_n$ is a strictly positive Hausdorff moment sequence for $n \geq 1$, with a possible mass at zero that affects only E_1 . For every fixed $K \geq 0$,

$$E_n = \frac{1}{Hn^2} \left\{ \sum_{k=0}^K \frac{c_k(b)}{\log^{k+2} n} + O_K(\log^{-K-3} n) \right\}, \quad c_k(b) = (k+1)! [u^k] \frac{(1+u)e^{bu}}{\Gamma(1-u)}. \quad (8.2)$$

In particular $E_n \sim 1/(Hn^2 \log^2 n)$.

Proof. The moment hypothesis implies $a_j = H \log j + O(1)$: subtract H inside the increment formula, use $h(t) - H = O(1 - t)$ near one, and sum the resulting $O(j^{-2})$ errors. Thus the defining root exists exactly as before. With $J(w) = w - c - \int h(t)/(w - t) dt$, the same operator has quadratic form

$$(c - I_0)|x|^2 + \int_0^1 th(t)|f(t) + x/t|^2 dt \geq 0.$$

The unique point of its spectrum above one is a simple root of J . Its interior spectral density is $h(t)/|J_+(t)|^2$; local C^1 regularity gives the continuous boundary values needed for inversion. There is no atom at one. If $c > I_0$ there is no atom at zero; if $c = I_0$, that atom has mass $(1 + \int h(t)/t^2 dt)^{-1}$, interpreted as zero for a divergent integral. The same coefficient comparison therefore gives a positive moment formula, with an additional atom at zero only when described.

The endpoint decomposition used in (3.2) now gives

$$\frac{\Re J_+(1-u)}{h(1-u)} = \log u + b + O(u \log(1/u)), \quad \frac{\Im J_+(1-u)}{h(1-u)} = \pi.$$

Consequently the density of the remainder measure is

$$\frac{u}{H} \left\{ \frac{1}{(\log u + b)^2 + \pi^2} + O\left(\frac{u}{1 + \log^2 u}\right) \right\}.$$

Changing to $y = nu$ gives, up to $O(n^{-3} \log^{-2} n)$ and exponentially small terms, the integral

$$\frac{1}{Hn^2} \int_0^\infty \frac{ye^{-y}}{(\log n - \log y - b)^2 + \pi^2} dy.$$

For completeness, on any fixed $0 < u \leq \delta < 1$,

$$|(1-u)^{n-1} - e^{-nu}| \leq C_\delta e^{-nu}(u + nu^2).$$

Integrating this bound against the leading density proves the stated error for the exponential-kernel replacement. The finite geometric expansion from Section 4 applies with b in place of γ_E . Its gamma-moment generator is $(1+u)e^{bu}/\Gamma(1-u)$, proving (8.2). The atom at zero contributes nothing when $n \geq 2$. $\square$

For example, replace Ford's coefficients by $a_j + \theta$, with $\theta \geq 0$. The kernel h stays fixed, while c increases by θ and $b = \gamma_E - \theta$. After subtracting the new pole term, the remainder is

$$E_n^{(\theta)} = \frac{1}{n^2 \log^2 n} \left[1 + \frac{2(1-\theta)}{\log n} + \frac{3\theta^2 - 6\theta - \pi^2/2}{\log^2 n} + O_\theta(\log^{-3} n) \right]. \quad (8.3)$$

At $\theta = 1$ the first inverse-logarithmic correction vanishes. This illustrates which part of the error is universal (the scale and leading constant) and which part records the detailed kernel and initial coefficient.

9 Reproducible computations and numerical behavior

The analytic proof has three independently evaluable descriptions: the convolution recurrence, the spectral measure of a positive operator, and the endpoint Laplace integral. The supplied experiments compare them. None of the numerical calculations is used as a substitute for an infinite-dimensional argument or an asymptotic remainder proof.

9.1 Precision and independent representations

The principal run computes the recurrence through $n = 1000$ using 370 decimal digits. It propagates $u_n = \rho^n g_n$ so that large intermediate values are avoided, and then recovers $E_n = \rho^{-n}(\gamma - u_n)$. Since this subtraction loses about $n \log_{10}(1/\rho)$ digits, ordinary machine precision would be unsuitable. A higher-precision comparison is included in the run log.

An independent calculation replaces the integral in J by Gauss–Legendre quadrature. This gives a finite symmetric matrix with one central coordinate and diagonal entries at the quadrature nodes. Its spectral weights approximate the continuous positive measure. We used 128, 256, and 512 nodes. At the largest resolution, the worst sampled relative discrepancy in the residual is about 1.1×10^{-8} . This discretization evaluates the small residual from positive spectral weights and does not subtract the dominant exponential term.

A third calculation uses the exact local identity for $P(s)$ and rescales the integral by $y = ns$. The recorded run uses 70 decimal digits, a local series of degree 24, and a scaled cutoff of 120. A sensitivity run uses 90 digits, degree 32, and cutoff 160. For that stronger run, agreement with the recurrence has relative discrepancy below 5×10^{-44} at $n = 50$ and below 4×10^{-66} at $n = 256, 1000$. Those figures describe agreement between numerical calculations; they are not directed-rounding error certificates.

The scripts also check the exact symbolic coefficient identities, the Taylor coefficients of the algebraic expansion, and the finite geometric identity underlying the product bounds. The stored floating diagnostics record positive E_n for all computed indices and positive alternating differences at sampled orders through six. Strict positivity for every index and order follows from the proof, not from this finite sample.

Table 1: Normalized residual from the scaled positive integral. Values are floating diagnostics, rounded to ten decimal places.

n	$n^2(\log n)^2 E_n$
10^2	0.9692664545
10^3	1.0978392280
10^4	1.1195611911
10^6	1.1070852241
10^{10}	1.0750315447
10^{20}	1.0407938318
10^{50}	1.0169800175
10^{100}	1.0085903949

9.2 Why the leading asymptotic can be misleading at first

Table 1 illustrates slow logarithmic convergence. The normalized quantity passes through one and overshoots before returning toward its limit. This is compatible with the eventual positive first correction $2/\log n$. Complete monotonicity applies to E_n , not to the separately normalized sequence $n^2(\log n)^2 E_n$.

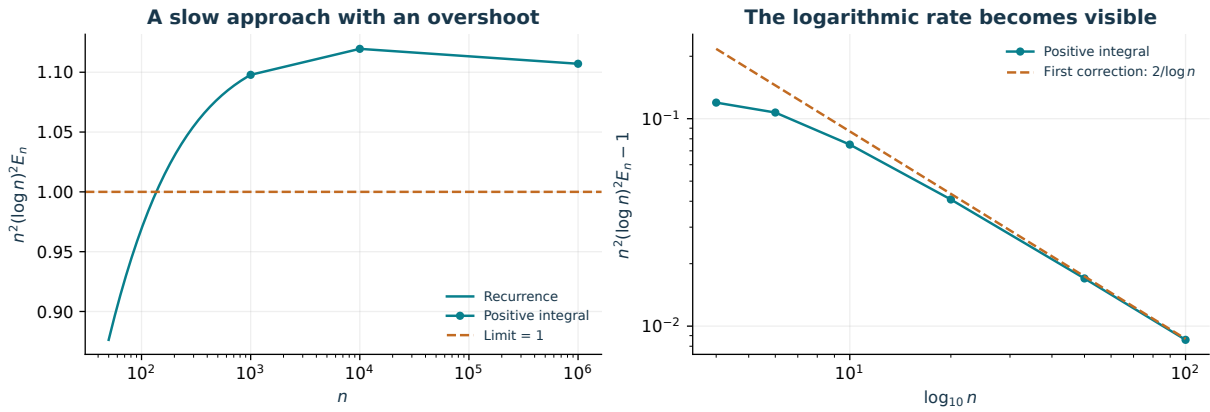


Figure 1: The finite-index approach and the eventual logarithmic rate. The right horizontal axis is $\log_{10} n$, itself displayed on a logarithmic scale. The curve $2/\log n$ is the first proved relative correction, not a fitted model.

Nor does a longer fixed inverse-logarithmic truncation always improve a numerical approximation. For example, at $n = 1000$, including $c_0, \dots, c_3$ gives relative error approximately -7.17×10^{-4} , whereas including c_4 as well gives approximately -1.67×10^{-2} . Asymptotic expansions assert improvement as the limiting parameter tends to infinity at fixed order, not a pointwise monotonic improvement with the order.

9.3 Accuracy when the logarithmic integrals are retained

Define the three successive approximations

$$\begin{aligned}
 A_0(n) &= n^{-2} F_2(\log n), \\
 A_1(n) &= A_0(n) + (2n^3)^{-1} F_3(\log n), \\
 A_2(n) &= A_1(n) + n^{-4} \{F_4(\log n)/12 - p_2 F_4'(\log n)\}.
 \end{aligned}$$

Their errors, computed against the positive endpoint integral, are shown in Table 2. The signs are retained in the table; the plot uses absolute relative error. The next proved coefficient in (1.13) explains the remaining error of A_2 .

Table 2: Signed relative errors $A_j(n)/E_n - 1$.

n	A_0	A_1	A_2
256	-4.472×10^{-3}	-5.494×10^{-7}	9.520×10^{-8}
500	-2.280×10^{-3}	-2.330×10^{-7}	1.230×10^{-8}
1000	-1.133×10^{-3}	-8.804×10^{-8}	1.442×10^{-9}

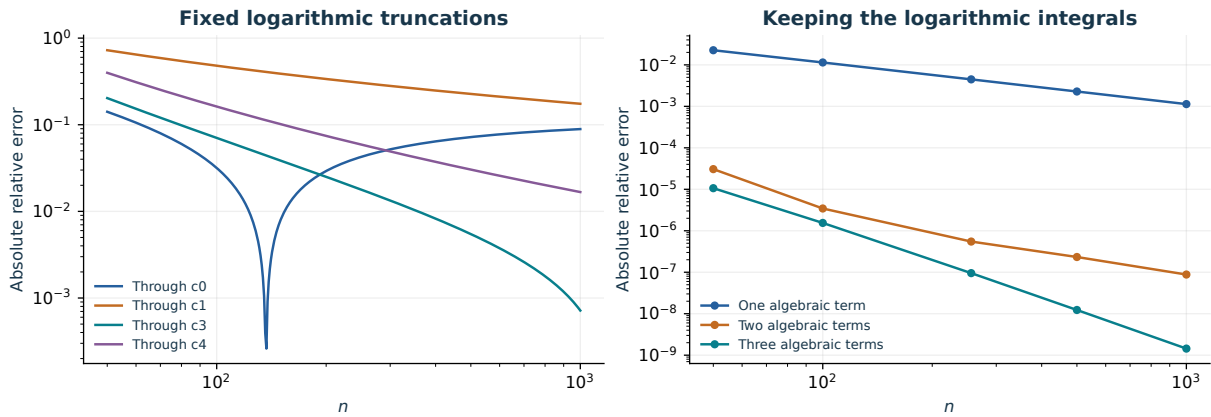


Figure 2: Two types of approximation. On the left, a fixed number of logarithmic terms may suffer cancellation or grow inaccurate at moderate indices. On the right, the exact logarithmic integrals are retained while successive powers of $1/n$ are added.

9.4 The product constant

The recorded finite product at $m = 1000$ is

$$P_{1000} = 0.61788133434828841381086077163828710263 \dots$$

The theorem proves that it lies above C_* , and (5.11) provides an explicit tail enclosure in exact arithmetic. The displayed decimal is a high-precision numerical estimate of the finite product, rather than a certified decimal enclosure of the infinite one.

Table 3: Selected normalized partial products P_m , rounded by truncating the displayed decimal expansion.

m	P_m
1	0.6480070859547479028130
2	0.6244817040471193873375
5	0.6181251627013800807835
10	0.6178845005447616555349
50	0.6178813343482884163074
1000	0.6178813343482884138108

9.5 Reproduction commands and scope

The Python programs require `mpmath`, `sympy`, `numpy`, `scipy`, and `matplotlib`. Versions used in preparation are recorded in the data. From the package root:

```
python code/verify_recurrence.py --nmax 1000 --out rerun/data
python code/verify_symbolic.py --out rerun/symbolic_checks.json
python code/make_figures.py --input rerun/data --out rerun
make pdf
```

The first three commands keep the originally supplied data intact. Compilation uses the already included figures and tables, so building the PDF does not require numerical libraries or network access. Normal Python and optimized `python -O` runs produced the same deterministic numerical data files in the preparation checks. Runtime measurements and metadata are not expected to be identical across machines.

10 Further research questions and proposed next steps

The exact measure separates several questions that were difficult to distinguish from the original recurrence. The following are concrete extensions, with their unresolved parts stated explicitly.

1. **Complete the priority comparison.** Obtain the full Ford–Lau solution and inspect subsequent work on this specific recurrence, including citations of the original problem. Determine precisely which of the density formula, complete monotonicity, all-orders coefficients, and product correction have appeared before. This is the first step before describing the result as a first publication.
2. **Optimal truncation of the inverse-logarithmic expansion.** The coefficient generator is an entire reciprocal-gamma expression, but the asymptotic coefficients also contain $(k+1)!$. Determine their large-order behavior, the optimal truncation index as a function of $\log n$, and the least attainable remainder for this single series. The exact F_a integrals provide a reference against which a proposed optimal-truncation theorem can be tested. Fixed-order error estimates in this paper do not answer this question.
3. **Uniformity when the algebraic order grows.** Theorem 1.3 and its general version hold for each fixed truncation order. Quantify the dependence of the remainder constant on that order, and determine a range $R = R(n) \rightarrow \infty$. The singularities at distance 2π in the endpoint correction P are natural inputs, but the logarithmic denominator must be controlled simultaneously.
4. **Interval-certified product constants and errors.** Implement directed-rounding enclosures for ρ , γ , and the positive integral. Combine them with (5.11) or (5.12) to certify the infinite product C_* to an arbitrary requested number of digits. An implementation should avoid subtraction of two exponentially large terms. The current scripts demonstrate numerical agreement; they do not provide these interval guarantees.
5. **Growing finite-difference order.** The exact formula for $D_{n,r}$ remains valid for every r , whereas (7.1) assumes fixed r . When r/n stays away from zero and infinity, its integral is controlled by an interior beta-distribution saddle. Derive a uniform theorem joining that interior regime to the logarithmic endpoint regime $r = o(n)$. Such a theorem would describe the full array of difference inequalities, not just each fixed column.
6. **General logarithmic renewal kernels.** Suppose $a_{j+1} - a_j = \int_0^1 t^{j-1} h(t) dt$ for a positive density h , and $1 + a_1 \geq \int_0^1 h(t)/t dt$. Theorem 8.1 gives an all-orders logarithmic expansion

under explicit regularity assumptions, with a possible atom at zero. Determine minimal endpoint regularity for that conclusion, and characterize precisely which deformations preserve the special cancellation of Euler's constant in (1.6). If $h(1) > 0$ and a suitable logarithmic endpoint limit exists, the same calculation predicts leading constant $1/h(1)$.

7. **Stability under perturbing initial coefficients.** A finite change of the recurrence coefficients leaves their logarithmic tail unchanged but may create additional poles. Classify the perturbations for which the positive remainder measure survives, and those for which oscillating exponential corrections appear. The singularity at one should still determine a universal logarithmic component after all relevant poles are removed; that assertion requires a separate global zero analysis.
8. **Arithmetic error propagation.** Insert the proved product correction into a fully audited totient-volume argument and track every competing approximation. Determine whether any arithmetic regime exposes the new $\rho^m/(m^2 \log^2 m)$ scale rather than hiding it under a larger sieve or boundary error. This paper proves the continuous-scale correction and leaves that arithmetic comparison open.
9. **Lean formalization in stages.** Start with the convolution recurrence, the integral formula for its increments, and the algebraic relation between G and J . Next formalize the positive bounded operator and its scalar spectral measure, including endpoint atoms. The exact moment inequalities then follow with little asymptotic machinery. A final stage can formalize the uniform Laplace estimates and coefficient extraction. No declaration in the supplied package is represented as a kernel-checked proof.

11 Integration and reproducibility record

The package is self-contained for compilation and for the supplied numerical experiments. It contains the complete LaTeX source, a compiled PDF, vector figures and generated tables, Python programs, CSV/JSON data, a build file, and a claim-status ledger. No original repository manuscript or third-party PDF is redistributed.

The inspected repository snapshots were

Repository	Commit
openai/math	adc7f1241b42e322a6451854ab7e4b4c146bf78a
ProveIt	6d78805163d6b5b92d27f8f41555254c49fe0963

The first snapshot supplies the recurrence and product-scale anchor; the second records the repository scope at the start of the study. Their full paths and the relevant source blob identifier appear in `provenance.json`. The inspection was targeted, and did not constitute a theorem-by-theorem audit of either repository.

An appropriate proposed destination is `Analysis/RenewalTheory/Research/Ford_Renewal_Logarithmic_Remainder/`. This is an integration suggestion; no existing repository content has been changed or published by this package.

The mathematical dependency is short: a positive operator gives the exact residual measure; elementary endpoint estimates give its sharp size and explicit bounds; a convergent polylogarithm identity gives the complete expansions; positive integral identities transport them to products and renewal accuracy. The numerical programs check each description against an independently computed representation. Their purpose is reproducibility and error detection, while the proofs above establish the infinite statements.

References

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- [4] G. Teschl, *Topics in Linear and Nonlinear Functional Analysis*, author's textbook page and contents, especially the spectral theorem and spectral measures in Chapter 5. <https://www.mat.univie.ac.at/~gerald/ftp/book-fa/fa.html>. The general bounded self-adjoint spectral theorem is the standard functional-analytic input used in Section 2.
- [5] OpenAI, *An asymptotic formula for the number of totients*, repository manuscript dated 25 September 2026, in `openai/math`. Inspected source: `preprints/An-asymptotic-formula-for-the-number-of-totients-September-25-2026/build/source/statements.tex`, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. <https://github.com/openai/math>. Only the explicitly defined recurrence and product scale are used here; the manuscript's main arithmetic theorem is not an input.
- [6] V. Reshetnikov, *ProveIt*, research and formal-mathematics repository, snapshot `6d78805163d6b5b92d27f8f41555254c49fe0963`. <https://github.com/VladimirReshetnikov/ProveIt>. The inspected asymptotic and conditioning material was used to select a nonduplicative direction, not as an unproved lemma in this article.